

Positive distribution-function modelling of ω Centauri

AGAMA infrastructure, numerical validation and first fitting pilot

oCen_dm project notes

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Abstract

We describe a spherical stellar distribution-function (DF) modelling branch built on AGAMA. Positive analytic action DFs, or positive mixtures of these DFs, generate the stellar density and velocity moments in a common potential. The stellar contribution to the potential is iterated to self-consistency; a central point mass, Plummer remnants and a truncated dark halo may also contribute. The same model predicts the photometric profile and 89 kinematic measurements, preserving the selection-aware bin averages used in the Jeans analysis. Four example equilibria pass independent projection and complete resolution-doubling checks, with predicted dispersions changing by at most 0.011 observational errors. A positive mixture produces radial anisotropy at intermediate radii and tangential anisotropy farther out. A short no-DM pilot exercises the joint fitting workflow and reaches $\chi^2_{\text{kin}} = 761.52$, but stops at its 96-evaluation budget before optimizer convergence. These results establish a working and tested inference infrastructure. They do not yet measure new dark-matter or black-hole constraints. The DF guarantee applies to the stars; the remnant and dark-matter components are presently specified as static gravitational sources.

1 Purpose and scope

The Jeans modelling sequence established diagnostic baselines for the kinematic data. A Jeans solution specifies density, anisotropy and velocity moments, but positive density and positive second moments alone do not establish the existence of a non-negative phase-space DF. The new branch starts with a positive stellar DF and derives those quantities together. It therefore tests whether the observed photometry and kinematics can be described by a realizable equilibrium within a stated family.

AGAMA supplies the action calculations, analytic DF implementation and gravitational-potential machinery [1, 2]. Our Python layer controls the spherical velocity integrals, the self-consistency iteration, projection, data likelihood and run provenance. This branch is separate from the existing `AgamaDFBackend`, which uses a restricted `QuasiSpherical` inversion as a diagnostic. No inverted negative DF is clipped in the new construction.

The current model is spherical, time independent and nonrotating in its even DF. The luminous components share one stellar mass-to-light ratio. Density and anisotropy can change together, including the unresolved central density in the presence of a black hole. We retain the previous Jeans results as benchmarks; this infrastructure does not certify or reweight their saved posterior samples.

2 Constructive stellar distribution functions

2.1 Actions and the single-component family

For a spherical potential with $\Phi(\infty) = 0$, the energy and angular momentum of a bound star are

$$E = \frac{1}{2}|\mathbf{v}|^2 + \Phi(r) < 0, \quad L = |\mathbf{r} \times \mathbf{v}|. \quad (1)$$

The radial action is

$$J_r(E, L) = \frac{1}{\pi} \int_{r_{\text{peri}}}^{r_{\text{apo}}} \left[2\{E - \Phi(r)\} - \frac{L^2}{r^2} \right]^{1/2} dr. \quad (2)$$

AGAMA's action coordinates satisfy $L = J_z + |J_\phi|$ in the spherical case. We use its `DoublePowerLaw` family, with rotation, the optional core modifier and the optional exponential action cutoff disabled:

$$f_i(J_r, L) = C_i \left[1 + \left(\frac{J_{0,i}}{h_i} \right)^{\eta_i} \right]^{\Gamma_i/\eta_i} \left[1 + \left(\frac{g_i}{J_{0,i}} \right)^{\eta_i} \right]^{-B_i/\eta_i}, \quad (3)$$

$$h_i = h_{r,i} J_r + \frac{3 - h_{r,i}}{2} L, \quad g_i = g_{r,i} J_r + \frac{3 - g_{r,i}}{2} L. \quad (4)$$

Equal coefficients of J_z and $|J_\phi|$ enforce spherical symmetry. The coefficients h_r and g_r control relative orbital weights; they are not values of the velocity-anisotropy parameter β .

The parameter domain is $J_0 > 0$, $\eta > 0$, $\Gamma < 3$, $B > 3$ and $0 < h_r, g_r < 3$. It makes equation (3) non-negative throughout bound action space and gives finite total mass. Negative Γ is allowed and suppresses the DF at small actions. At the isolated zero-action limit the DF may vanish or diverge while remaining integrable. Unbound phase space has $f = 0$. The point $r = 0$ itself is excluded from numerical evaluation when the potential contains an uns softened point-mass singularity.

AGAMA determines C_i using the requested component `mass`, so that

$$(2\pi)^3 \int f_i(\mathbf{J}) d^3J = M_i. \quad (5)$$

The integral uses $J_r, J_z \geq 0$ and signed J_ϕ . The library's `norm` parameter instead sets a prefactor whose relation to total mass depends on DF shape; it is not used here. Internal units are pc, $M_\odot$, km s^{-1} and pc km s^{-1} for actions.

2.2 Positive mixtures and emergent anisotropy

For K components, we set $M_i = w_i M_\star$ and

$$f_\star = \sum_{i=0}^{K-1} f_i, \quad w_i > 0, \quad \sum_i w_i = 1. \quad (6)$$

These are basis components of the same observed stellar population. A common mass-to-light ratio makes their mass and light fractions equal. The optimizer can vary $a_i = \ln(w_i/w_0)$ for $i \geq 1$, with $a_0 = 0$, and recover

$$w_i = \frac{\exp(a_i)}{\sum_j \exp(a_j)}. \quad (7)$$

The numerical implementation subtracts the largest a_i before exponentiating. This parameterization preserves positivity and normalization without fitting raw fractions independently.

The density and pressures follow from the DF:

$$\rho_\star = \int f_\star d^3v, \quad p_r = \int v_r^2 f_\star d^3v, \quad p_t = \frac{1}{2} \int (v_\theta^2 + v_\phi^2) f_\star d^3v, \quad \beta(r) = 1 - \frac{p_t}{p_r}. \quad (8)$$

Here p_t refers to one tangential component. For the nonrotating DF, $\sigma_r^2 = p_r/\rho_\star$ and $\sigma_\theta^2 = \sigma_\phi^2 = p_t/\rho_\star$. Neither $\rho_\star(r)$ nor $\beta(r)$ is imposed independently. Mixtures with different action scales and orbital weights can produce an anisotropy turnover as their contributions to the pressure change with radius.

Figure 1 demonstrates this flexibility for the example configurations in Table 1. The two-component example has $\beta \simeq +0.14$ at 5 pc, $+0.04$ at 20 pc and -1.0 at 50 pc. These are example equilibria, not inferred anisotropy profiles.

Table 1: Fixed-parameter examples used for infrastructure validation. All have $M_\star = 2.7 \times 10^6 M_\odot$, $M_\bullet = 10^4 M_\odot$, $M_{\text{rem}} = 5 \times 10^5 M_\odot$, $a_{\text{rem}} = 5.3 \text{ pc}$ and $D = 5.43 \text{ kpc}$. Every DF has $\Gamma = 0$, $B = 7$, $\eta = 2$ and $h_r = 1.5$. Halo examples have $r_s = 30 \text{ pc}$ and $r_t = 1000 \text{ pc}$.

Example	$M_{\text{DM}}(< 100 \text{ pc})$ [$M_\odot$]	γ	w_i	$J_{0,i}$ [pc km s^{-1}]	$g_{r,i}$
Single DF, no DM	0	—	1	300	1.0
Single DF, core	2×10^6	0	1	300	1.0
Single DF, cusp	2×10^6	1	1	300	1.0
Two DFs, no DM	0	—	0.75, 0.25	260, 600	1.1, 2.0

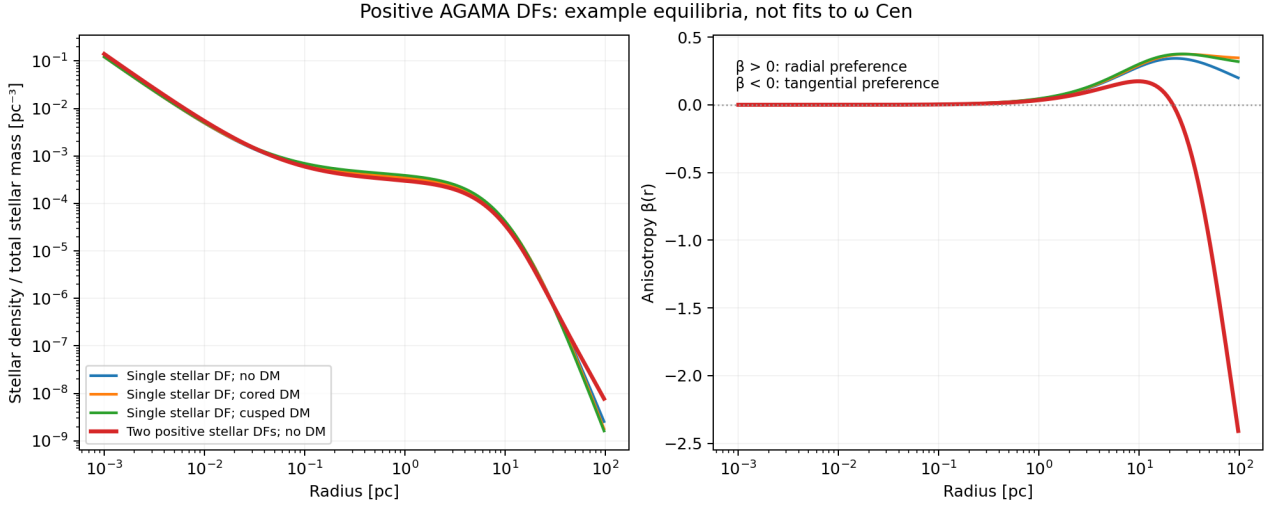


Figure 1: Stellar density per unit total stellar mass (left) and derived anisotropy (right) for four positive-DF example equilibria. All use an exact central point mass; their inner stellar density is an output of the DF. The two-component model changes from radial preference at intermediate radii to tangential preference farther out. The examples demonstrate realizable profile shapes and have not been optimized against the observations.

3 Gravitational model and self-consistency

3.1 Live stellar gravity and static components

The total potential is

$$\Phi = \Phi_\star[f_\star] + \Phi_\bullet + \Phi_{\text{rem}} + \Phi_{\text{DM}}, \quad \Phi_\bullet = -\frac{GM_\bullet}{r}, \quad \Phi_{\text{rem}} = -\frac{GM_{\text{rem}}}{\sqrt{r^2 + a_{\text{rem}}^2}}. \quad (9)$$

AGAMA’s Plummer potential with zero scale radius implements the exact point mass; no BH softening is introduced. The optional dark halo follows the same truncated generalized NFW law as the Jeans branch:

$$\rho_{\text{DM}}(r) = \rho_s \left(\frac{r}{r_s}\right)^{-\gamma} \left(1 + \frac{r}{r_s}\right)^{\gamma-3} \left[1 + \left(\frac{r}{r_t}\right)^2\right]^{-2}. \quad (10)$$

The code fixes ρ_s from $M_{\text{DM}}(< 100 \text{ pc})$ and constructs a spherical multipole potential. The supplied halo examples use $\gamma = 0$ or 1.

Only stellar gravity responds to the DF iteration. The remnant and halo densities are non-negative gravitational sources but have no specified DFs in this branch. Consequently, stellar phase-space consistency is established within the numerical tolerances, while a fully live multi-component phase-space model remains a later extension. Positive stellar DFs also do not, by themselves, prove dynamical stability or survival in the Galactic tide.

3.2 Velocity quadrature and the iteration

For an even spherical DF, let $\mu = |v_r|/v \in [0, 1]$ and $v_{\text{esc}}(r) = \sqrt{-2\Phi(r)}$. We evaluate

$$\rho_\star(r) = 4\pi \int_0^{v_{\text{esc}}} v^2 dv \int_0^1 f_\star d\mu, \quad (11)$$

$$p_r(r) = 4\pi \int_0^{v_{\text{esc}}} v^4 dv \int_0^1 \mu^2 f_\star d\mu, \quad (12)$$

$$p_t(r) = 2\pi \int_0^{v_{\text{esc}}} v^4 dv \int_0^1 (1 - \mu^2) f_\star d\mu. \quad (13)$$

AGAMA evaluates the actions and analytic DF at each quadrature point. Positive Gauss–Legendre weights preserve the sign of these integrals.

The iteration begins with a stellar Plummer seed of the requested stellar mass and a 5-pc scale radius, plus the static components. At each iteration we integrate the DF over velocities, interpolate $\ln \rho_\star$ against $\ln r$, construct the new stellar multipole potential, and recompute the actions in the total potential. The density interpolation uses a cubic spline with explicit power-law continuation at both ends; unacceptable finite-mass asymptotes are rejected. The seed is a numerical starting point, not the stellar density retained in the final model.

Both density and total force must converge. Testing the force alone could hide a poorly converged stellar density when a static component dominates the potential. The default stopping criterion is

$$\max_{r_j} \left\{ \left| \frac{\rho_\star^{(n)}(r_j)}{\rho_\star^{(n-1)}(r_j)} - 1 \right|, \left| \frac{F^{(n)}(r_j)}{F^{(n-1)}(r_j)} - 1 \right| \right\} < 5 \times 10^{-4}, \quad (14)$$

within 50 iterations. A fresh density integral in the final potential then uses twice the velocity quadrature order at geometric midpoints between radial grid nodes. The maximum discrepancy from the interpolated density must be below 0.5%, as must the error in the total stellar mass inferred from its potential. Failure raises a numerical rejection; the code does not replace negative or non-finite values with a floor.

An early prototype using AGAMA’s standard `SelfConsistentModel` helper showed a discrepancy of approximately 11% near 0.008 pc between its density representation and a fresh DF integral. Independent spherical velocity quadrature confirmed that discrepancy. The explicit iteration above avoids it in the tested models and exposes the integration resolution for checking. This observation concerns the tested configuration; it is not a general diagnosis of AGAMA or a modification to the installed library.

Table 2: Default numerical settings. The final validation rebuild doubles all four node counts and halves the iteration tolerance. The radial domain is unchanged in that resolution test.

Quantity	Default
Radial integration domain	10^{-4} – 10^5 pc
Spherical potential/density nodes	160
Velocity nodes in each of v/v_{esc} and μ	64
Intrinsic density/pressure interpolation nodes	240
Line-of-sight quadrature nodes	128
Maximum iterations	50
Iteration tolerance	5×10^{-4}
Maximum density and mass closure errors	5×10^{-3}

4 From the DF to the observations

4.1 Projection and binning

The cached density and pressure tables are interpolated in logarithms using shape-preserving piecewise cubic interpolation. With $q = R^2/r^2$, the projected surface density and unnormalized second moments are

$$\Sigma(R) = 2 \int_R^\infty \frac{\rho_\star(r) r \, dr}{\sqrt{r^2 - R^2}}, \quad (15)$$

$$P_{\text{los}}(R) = 2 \int_R^\infty \frac{[(1-q)p_r + qp_t]r \, dr}{\sqrt{r^2 - R^2}}, \quad (16)$$

$$P_{\text{pmR}}(R) = 2 \int_R^\infty \frac{[qp_r + (1-q)p_t]r \, dr}{\sqrt{r^2 - R^2}}, \quad (17)$$

$$P_{\text{pmT}}(R) = 2 \int_R^\infty \frac{p_t r \, dr}{\sqrt{r^2 - R^2}}. \quad (18)$$

The substitution $r = R \cosh u$ removes the integrable endpoint singularity. Numerically the upper limit is the radial integration boundary, 10^5 pc; requests beyond one hundredth of that radius require a wider grid. These are projections of DF-derived moments. The Jeans equation is used as an independent test, not as a second moment solver in this branch.

The ratio P_k/Σ is averaged using the selection representation saved with each observed profile. HST and Gaia component bins use equal-weight radial quantiles of the measured stars. For profiles with annular edges and no stored stellar quantiles, six Gauss–Legendre nodes implement the tracer-weighted average

$$\overline{v_{kb}^2} = \frac{\int_{R_{b,-}}^{R_{b,+}} P_k(R) R \, dR}{\int_{R_{b,-}}^{R_{b,+}} \Sigma(R) R \, dR}. \quad (19)$$

A profile without edges uses its representative radius. Proper-motion second moments are divided by $(4.74047046D)^2$ for D in kpc, giving $(\text{mas yr}^{-1})^2$. Angular radii become physical radii through $R = r_{\text{arcsec}}(1000D)/206264.806$.

4.2 Streaming and the kinematic likelihood

For continuity with the baseline data treatment, the predicted dispersion in a bin is

$$s_b = \left(\overline{v_{kb}^2} - S_b \right)^{1/2}, \quad (20)$$

where S_b is the saved published streaming variance in the corresponding observational units. A non-positive radicand is rejected. This is a moment-level approximation to rotation. We have not fitted the odd part of a rotating DF, and positivity of the even DF does not establish that a positive rotating extension can reproduce every adopted streaming moment.

The five profiles contain 21 HST radial, 21 HST tangential, 29 MUSE LOS, nine Gaia radial and nine Gaia tangential bins. For a measured dispersion d_b with lower and upper errors $e_{b,-}, e_{b,+}$, the normalized split-normal likelihood is

$$\ln \mathcal{L}_{\text{kin}} = \sum_b \left[-\frac{(s_b - d_b)^2}{2e_b^2} + \ln \frac{\sqrt{2/\pi}}{e_{b,-} + e_{b,+}} \right], \quad e_b = \begin{cases} e_{b,+}, & s_b > d_b, \\ e_{b,-}, & s_b \leq d_b. \end{cases} \quad (21)$$

The current likelihood is diagonal. Reusing the selected profiles and their error arrays does not establish that every bin or dataset is statistically independent. No instrument scale factors are fitted in the supplied examples.

4.3 Photometric shape and the joint objective

The composite profile combines inner HST star counts and outer surface brightness, with 82 retained points of positive relative weight. Under the common stellar mass-to-light ratio, write

$$m_j = -2.5 \log_{10} \Sigma(R_j), \quad \hat{A} = \frac{\sum_j (\mu_j - m_j) / \epsilon_j^2}{\sum_j 1 / \epsilon_j^2}, \quad \chi_{\text{phot}}^2 = \sum_j \frac{(m_j + \hat{A} - \mu_j)^2}{\epsilon_j^2}. \quad (22)$$

The common magnitude zero-point $\hat{A}$ is profiled out; photometry therefore constrains shape without fixing the stellar mass through an assumed absolute mass-to-light ratio.

The composite profile provides relative weights W_j , not calibrated photometric uncertainties. The pilot explicitly adopts $\epsilon_j = 0.1 / \sqrt{W_j}$ mag, with zero-weight points discarded. These errors define the diagnostic weighting and omit the covariance introduced by the profile splice. Users may supply explicit per-point errors in a photometric snapshot instead.

The local optimizer minimizes

$$Q = \chi_{\text{kin}}^2 + \chi_{\text{phot}}^2. \quad (23)$$

The normalized kinematic and profiled Gaussian photometric log-likelihoods are also saved. The search bounds are not a complete set of inference priors. Photometric error calibration, the zero-point treatment, distance and mixture priors need explicit choices before computing posterior probabilities or evidences. No DF-family evidence is reported here.

5 Numerical validation

The recorded verification pass contains 94 successful tests: 23 new DF tests and 71 existing consistency, anisotropy and run-replay tests. The new controls include positive-mixture normalization and spherical invariance, an analytic isotropic Kepler DF, exact point-mass gravity, the independent Jeans moment equation, radial integration of stellar mass, and halo normalization and force checks. They also exercise bound and unbound phase-space evaluation, unit protection, failed convergence, photometric normalization, bin averages and infeasible streaming subtraction. The installed library identifies itself as AGAMA 1.0.152, compiled on 8 June 2025.

For each full-data evaluation, we compare equation (18) with AGAMA’s native projected `GalaxyModel.moments` calculation at 16 radii from 0.02 to 80 pc. The maximum relative discrepancy among surface density and all three projected pressures must be below 0.5%. We then rebuild the equilibrium with the refined settings in Table 2. For each of the 89 kinematic bins we require

$$\delta_b = \frac{|s_b^{\text{refined}} - s_b^{\text{default}}|}{\min(e_{b,-}, e_{b,+})} < 0.1. \quad (24)$$

These checks measure numerical accuracy, separately from agreement with data. They do not prove convergence for every possible parameter choice. Particularly extreme DFs may need a larger domain or finer quadrature.

All four example equilibria and the pilot pass these checks (Figure 2). The initial single-DF no-DM example is closest to the native-projection tolerance, at 0.492%; its complete resolution test changes the binned dispersions by only 0.000605 data errors. The tests share AGAMA’s action and potential machinery, although their integration routes differ. The analytic Kepler control adds a reference that does not depend on the general action calculation for its expected density and pressure.

Table 3: Measured validation errors. Density closure uses a fresh DF integral at intermediate radii and doubled velocity order. Mass closure compares the stellar potential’s total mass with its requested DF mass. Projection is the largest discrepancy from native AGAMA surface-density/pressure integrals. The last column is the largest refined dispersion shift in data-error units.

Model	Density [%]	Mass [%]	Projection [%]	max δ_b
Single DF, no DM	0.02879	0.000439	0.49196	0.000605
Single DF, core	0.05801	0.000386	0.11473	0.008531
Single DF, cusp	0.04631	0.000434	0.03209	0.000664
Two DFs, no DM	0.04490	0.000204	0.05698	0.010868
No-DM pilot	0.03578	0.000630	0.06470	0.008509
Acceptance limit	0.5	0.5	0.5	0.1

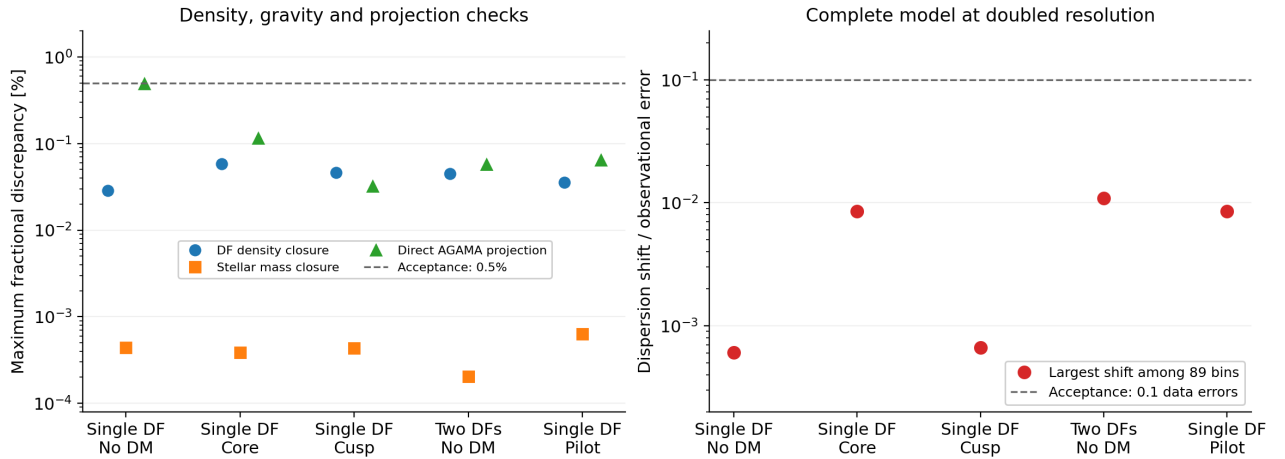


Figure 2: Numerical checks for the four fixed-parameter examples and the bounded no-DM pilot. Left: maximum fractional density, mass and direct projection discrepancies; the dashed line marks 0.5%. Right: largest change in a predicted dispersion when the entire calculation is rebuilt at doubled resolution, divided by the smaller quoted error of that bin. The dashed line marks 0.1 observational errors. Passing these tests establishes numerical control for these configurations, not a good fit.

6 First bounded fitting pilot

The pilot starts from the single-component no-DM configuration in Table 1. Bounded Powell optimization varies the six coordinates in Table 4; the remnant component, distance, η and h_r remain fixed. The example configuration files for DM models likewise leave the halo parameters fixed unless additional coordinates are explicitly supplied. These evaluations are not completed no-DM/DM comparisons.

Table 4: Search coordinates and the best joint model visited in the 96-call no-DM pilot. Logarithmic coordinates describe the search transformation; the intervals are optimizer bounds, not posterior priors. No uncertainties are implied. Fixed values are $M_{\text{rem}} = 5 \times 10^5 M_\odot$, $a_{\text{rem}} = 5.3 \text{ pc}$, $D = 5.43 \text{ kpc}$, $\eta = 2$ and $h_r = 1.5$.

Parameter	Coordinate	Search interval	Pilot value
$M_\star [M_\odot]$	$\ln M_\star$	$10^6 - 6 \times 10^6$	3.13958×10^6
$J_0 [\text{pc km s}^{-1}]$	$\ln J_0$	80–1000	293.094
B	Linear	4.5–10	6.99359
g_r	Linear	0.5–2.4	1.16588
Γ	Linear	−2–1.5	0.163119
$M_\bullet [M_\odot]$	$\ln M_\bullet$	$10^2 - 10^5$	12356.2

After 96 objective evaluations, the best joint objective falls from 17 972.67 to 1423.82. Its kinematic contribution falls from 15 881.77 to 761.52, and its photometric contribution from 2090.90 to 662.30.

The recorded elapsed time is 294 s, including validation and report generation. The history contains two numerically rejected trials. Figure 3 shows the accepted objective values and the best joint model retained after each call. The separate contributions need not decrease on every improvement in their sum.

The optimizer reached its evaluation limit before convergence. The residuals in Figure 4 remain substantial, particularly in the inner HST and photometric profiles. Table 5 gives the contributions to the kinematic objective. This pilot validates the fitting workflow; it does not measure the fit penalty imposed by DF positivity, establish the best attainable fit of the family, or yield a new BH or DM constraint. We do not assign a reduced chi-squared interpretation to the joint objective with these adopted photometric errors.

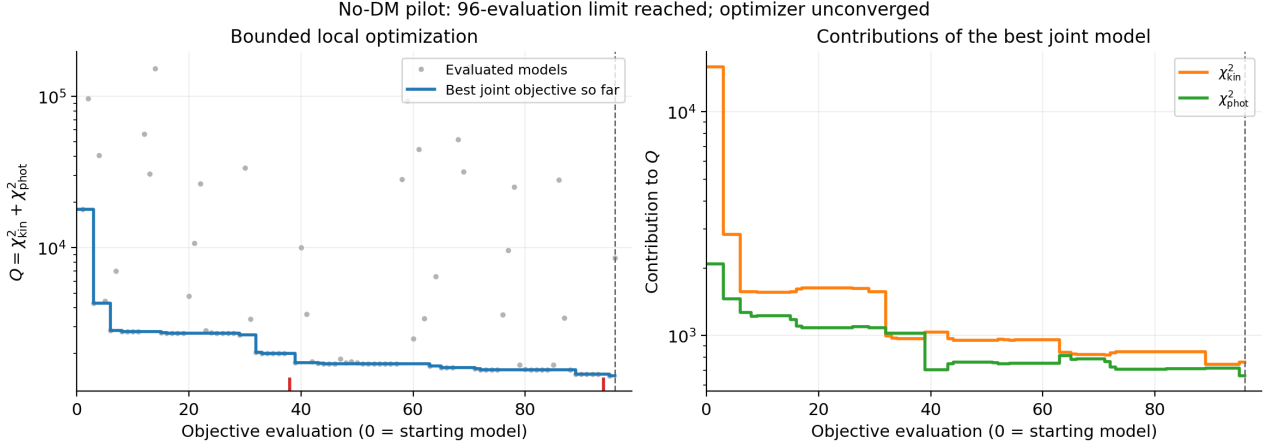


Figure 3: Recorded progress of the bounded no-DM pilot. Left: grey points are finite objective evaluations and the blue step curve is the lowest joint objective visited so far. Short red ticks at the bottom mark rejected numerical trials. Right: kinematic and photometric contributions of that same best joint model, rather than independent minima of each term. The vertical dashed line marks the imposed 96-call limit. The optimizer has not converged.

Table 5: Objective contributions for the best joint model visited by the pilot.

Profile	Number of bins	χ^2 contribution
HST radial PM	21	313.09
HST tangential PM	21	281.63
MUSE LOS	29	90.15
Gaia radial PM	9	58.98
Gaia tangential PM	9	17.67
Kinematic total	89	761.52
Photometric shape (one profiled zero-point)	82	662.30
Joint objective		1423.82

AGAMA positive stellar DF — diagnostic pilot, not a posterior fit

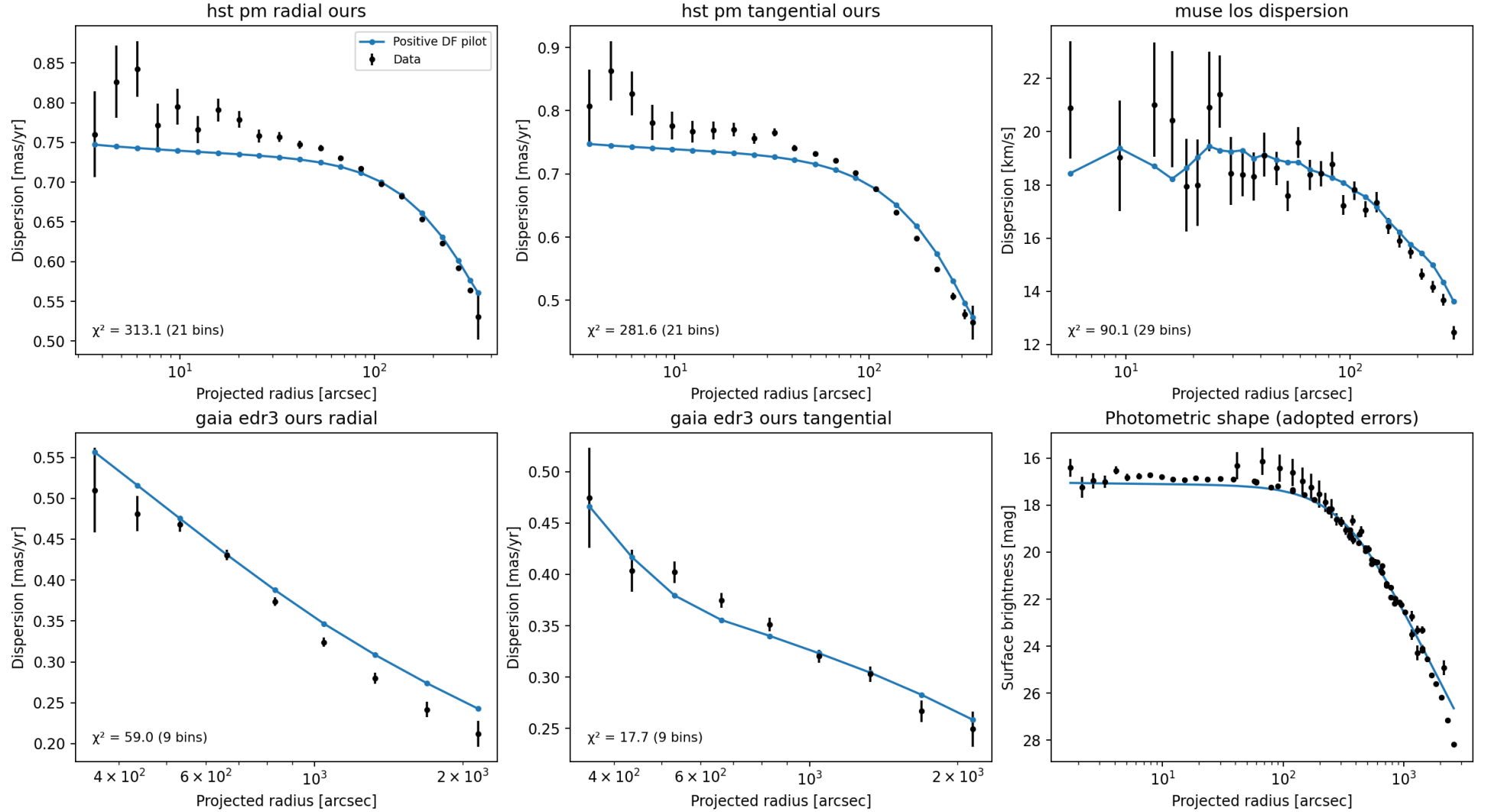


Figure 4: The bounded no-DM pilot against all five kinematic profiles and the composite photometric profile. Model points use each observed bin’s selection-aware second-moment average and saved streaming correction. The lines connect these bin predictions; they are not a continuously sampled intrinsic velocity-dispersion profile. Photometric bars show the adopted diagnostic errors, $0.1/\sqrt{W_j}$ mag. The common photometric zero-point is fitted. The clear remaining residuals require further optimization and model comparisons.

The pilot’s refined equilibrium changes its kinematic objective to 761.41 and its joint objective to 1423.76. Its largest dispersion shift is only 0.00851 data errors, so increasing numerical resolution does not remove the observed residuals. An independent fresh-process replay reproduces the saved objective, all 89 predictions and the photometric prediction exactly from the saved model and data.

7 Software interface, records and reproduction

Table 6: Main interfaces in the new branch. Both source modules reside in `src/ocen_dm/kinematics/`.

File or interface	Responsibility
<code>positive_df.py</code>	Validated component/configuration objects, positive DFs, self-consistent gravity, phase-space evaluation and moment projections.
<code>df_fit.py</code>	Explicit photometric data/errors, joint objective, bounded search coordinates and positive mixture-weight transformations.
<code>bin/run_df_pilot.py</code>	Fixed-parameter evaluation or bounded local optimization, snapshots, numerical refinement and a data/model figure.
<code>tests/test_positive_df.py</code>	Physical, numerical and data-interface controls.
<code>configs/df/</code>	Single no-DM/core/cusp and two-component examples.

The driver reads the saved kinematic snapshot and verifies its checksum and data fingerprint. Each new run records the actual likelihood arrays, photometric errors, configuration, source hashes and selected source copies, evaluation history, best model, numerical diagnostics and status. Current driver records also contain the AGAMA version and extension-binary hash. The two-component search can vary the log mixture ratio, both action scales and orbital weights. AGAMA units are global: the wrapper initializes the project units once and rejects an incompatible existing unit system rather than silently changing units of objects already in memory.

A fresh output directory is required. From the project root, with the project environment active and AGAMA installed, a bounded pilot is:

```
python bin/run_df_pilot.py \
  --config configs/df/no_dm.json \
  --out results/df/my_no_dm_pilot \
  --sigma-mag 0.1 \
  --max-evaluations 96
```

Omitting `-max-evaluations` performs a fixed-parameter evaluation and the numerical checks. For data replay, use the options `-data-run` with the previous run directory and `-photometry-snapshot` with its `photometry.json` file. For model replay, place the saved `best_model.json` content under the `model` key in a driver configuration and omit optimization. The driver does not launch posterior sampling.

The records used in this note are under `results/df/validation_*_20260922/`, `results/df/pilot_no_dm_20260922/` and `results/df/replay_no_dm_20260922/`. The test and replay record is `results/diagnostics/df_validation/verification.json`. Figures 2 and 3 are reproduced by `bin/plot_df_writeup.py`, which reads saved results and launches no fits. Its input checksums and plotted arrays are in `results/plot_data/df_writeup_diagnostics.json`.

All figure exports are PNGs in `plots/`. Numerical tables and provenance remain outside that directory. This PDF and its LaTeX source both reside in `docs/`; build intermediates belong in `results/diagnostics/df_writeup/`.

8 Next modelling steps

The next experiment is a converged search from several starting points for single and mixed stellar DFs, followed by matched no-DM, core and cusp comparisons with the same photometric treatment. Varying the photometric error scale will show how the current weighting influences the inferred stellar density and orbital structure. The stellar/remnant mass split, distance treatment and mixture priors need to be specified before launching production posterior fits.

A later rotation model should fit an admissible odd DF contribution and mean velocities jointly, replacing the present streaming approximation. Axisymmetric structure, population-dependent tracer selections and stellar mass segregation are further model extensions. Assigning DFs to the remnants and DM would allow a fully live equilibrium and controlled N -body realizations. Those additions would connect the mass modelling more directly to the independent tidal-survival experiments. The present infrastructure establishes the constructive stellar DF and numerical controls needed for that progression.

References

- [1] Vasiliev, E. 2019, *AGAMA: Action-based galaxy modelling architecture*, MNRAS, 482, 1525. [doi:10.1093/mnras/sty2672](https://doi.org/10.1093/mnras/sty2672); [arXiv:1802.08239](https://arxiv.org/abs/1802.08239).
- [2] AGAMA reference manual, sections on spheroidal action DFs, moments and self-consistent models. Parameter conventions checked against the locally installed source and the [public manual source](#).